Our Correspondence School.

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No Royal Road.

Experience is a good school, but there are few holidays, and out of the bitterness of weary days of misdirected effort, there finally comes to most men a passable knowledge of their calling. To the thinking and studious mind there are shorter roads to knowledge than the hard path of experience, though there is no royal road. The railroad man who reads our pages from month to month will learn more in one year than he will learn in ten years of toil and trial by himself, because he gets the experience of others. The injector may serve as an illustration. We publish this month the garnered facts in regard to the working of an injector, and a close perusal and careful study of the subject as presented in our pages will help any intelligent railroad man to a good working knowledge of the intricacies of this ingenious mechanical contrivance.

Second Series of Questions.

Note.—These questions are on the injector and are numbered from 39 to 60. Those from 61 to 79 were on the lubricator and were published in May.

39. What is an injector?

A.—An injector is an apparatus in which a jet of steam condensed by water imparts to the latter its velocity, with the result that the final energy of the combined steam and water is greater than that at which the water would issue from the boiler. This difference of energy in favor of the jet passing through the injector enables it to lift the boiler check and enter the boiler.

40. In a general way, what are the two kinds of injectors?

A.—In a general way, injectors are known as "Single Tube" injectors, when they have a single set of nozzles, and as "Double Tube" injectors when they have two sets of nozzles; one of the latter kind has the function of lifting the feed water and delivering it to the forcing set, which latter imparts to the water sufficient velocity to cause it to enter the boiler.

41. What is the difference between a lifting and a non-lifting injector?

A.—A lifting injector is placed above the highest water level of the tank from which the feed water supply is taken, so that the injector has to lift the water up to its own level. A non-lifting injector is placed below the lowest level of the water of the tank from which the feed water is taken, and it flows to the injector by gravity.

42. What are the essential parts of an injector?

A.—The essential parts of injectors are, in the first place, the nozzles, which perform the function of delivering or forcing the water into the boiler, and, in the second place, the operating mechanism, such as the lifting valve, steam valve, water valve, etc.

43. How should an injector be started?

A.—In starting an injector, if it is a lifting one, the lifting valve should be opened first, and when the water appears at the overflow, the forcing valve of the injector should be opened gradually to its full extent. In starting a non-lifting injector the water should be admitted to the injector first, and when it appears at the overflow the steam valve should be opened gradually to its full extent.



TEN-WHEELER ON THE MAINE CENTRAL.

44. Give some of the common causes for failures of injectors to work.

A.—The most common causes for failure of injectors are the following: Leak in the suction pipe. Obstructed strainer or strainer of insufficient size. Liming up of the nozzles. Loose hose lining. Obstructions in the nozzles, such as pieces of coal, or other foreign matter washed in from the tank. Obstructions in the delivery pipe, such as a sticking boiler check which will not open properly. Leaky steam valve and boiler check, which will affect the starting of the injector by heating the suction pipe and the feed water.

45. What course should be pursued with check valve stuck open?

A.—In case the check valve is not provided with a stop valve, it will be necessary to close the heater cock and water valve of the injector, to prevent water from the boiler from running out through the injector. In this case reliance for feeding the boiler must be had on the injector, the check of which must be in good condition. If the boiler check has a stop valve, this can be closed down to shut off the boiler pressure from the check, in which case the check can be taken out for cleaning or for the removal of the causes which made the valve stick open.

46. How may it be determined whether the check valve or steam valve is leaking?

A.—To determine whether the check valve is leaking, the frost cock, with which all delivery pipes and most check valves are provided, should be opened. If water continues to issue from this frost cock, the indication is that the check valve is leaking. To determine whether the steam valve is leaking, the cap of the heater cock and the heater cock check should be removed. If the steam valve is leaking, steam will issue through the opening.

47. What may be done in this case?

A.—In such cases the check valve and the injector must be reported for repair, and the leaky valves ground in.

48. What may be done if a combining tube is obstructed?

A.—In case the combining tube is obstructed, it must be removed, the nozzles thoroughly cleaned, and all obstructions removed.

49. How may it be determined if the trouble is on account of a leak in the suction pipe?

A.—When the suction pipe leaks, the injector works with a hoarse, rumbling sound, caused by the air drawn in through the leaks. A leak in the suction pipe may also be determined by closing the tank valve, and opening the steam valve of the injector slightly, with the heater cock closed. If there is a leak anywhere in the suction line, the steam under such circumstances will issue through the leak.

50. What should be done in case of obstructed hose or strainer?

A.—In case of an obstructed hose or strainer, the connection between hose and strainer should be broken, and, with the heater cock closed, steam should be blown back through the strainer. The water allowed to flow through the open hose will usually wash out the obstruction. In most cases it will be sufficient to remove the waste cap of the strainer, and allow water from the tank to flow through to wash out the obstruction.

51. What should be done in case the feed water in the tank is too hot?

A.—In case the feed water in tank is too hot, it will be necessary to obtain

release his brakes independently of the leading engineer and without affecting any of the other brakes in the train, something that it is desirable to do when there is danger of overheating tires and in cases of wheel sliding. He can also apply the engine and tender brakes without applying the others on the train, independently of the head engineer.

To the distributing valve there are five distinct pipe connections: one from the engineer's automatic brake valve; one from the engineer's independent brake valve; one from the brake cylinder supply pipe branch of the return pipe; one from the driver, engine truck, and tender brake cylinders; and one from the brake pipe proper.

By means of these pipe connections the distributing valve is charged, the necessary variations in pressure made to operate the internal mechanism, and the supply of air conducted to the brake cylinders to apply the brake.

The internal construction of the various valves will be illustrated and described in a later number.

Beneath the automatic brake valve there is placed a double cut-out cock. This cut-out cock, when turned to closed position for the brake pipe, cuts out the automatic brake valve, but opens communication between the distributing valve and the independent brake valve by way of the pipe connection between them.

The duplex pump governor, shown in the diagram, is quite different from the older type of duplex governor.

This governor has one pressure top, which is the same as now used with the present standard, piped to the main reservoir direct, the connection being made to the return pipe at a point between the main reservoir and the main reservoir cut-out cock; and it controls the pump, when the brake valve is lapped, or it is necessary, for any reason, to close the main reservoir cut-out cock. The other, known as the excess pressure top, has two pipe connections, one to the brake pipe feed valve branch of the return pipe, made at any convenient point between the feed valve and the brake valve; and the other connection to the automatic brake valve, direct, and is supplied with air from the direct main reservoir port.

The construction of this pressure top is such that it provides for a differential pressure of twenty pounds between the brake pipe and the main reservoir pressures, regardless of what pressure the feed valve may be set at.

Between each of the brake cylinders and the distributing valve there is a cut-out cock, so that if any of these cylinders or their foundation brake gets out of order, they may be cut out independently of the others. In the brake pipe branch connection to the distributing valve there is a ⅜-in. cut-out cock for the purpose of cutting out simultaneously the engine and the tender brakes.

The brake cylinder hose couplings between the engine and the tender are of different dimensions from those of the brake pipe, so that they cannot be crossed when coupling them up.

The brake valves are mounted on permanent bases, and to these bases all the brake valve pipe connections are made, so that whenever necessary to remove the brake valves for cleaning or repairs no pipe joints need be disturbed.

The K feed valve is so constructed that, by turning the handle on the adjusting screw to one extreme position, main reservoir and brake pipe pressures of 90 and 70 lbs., respectively, are had; turning to the other extreme position, main reservoir and brake pipe pressures of 130 and 110 lbs., respectively, are had. By this arrangement but one feed valve is necessary either for the low pressure or for the high speed brake, and in addition, when using the ordinary 70-lb.

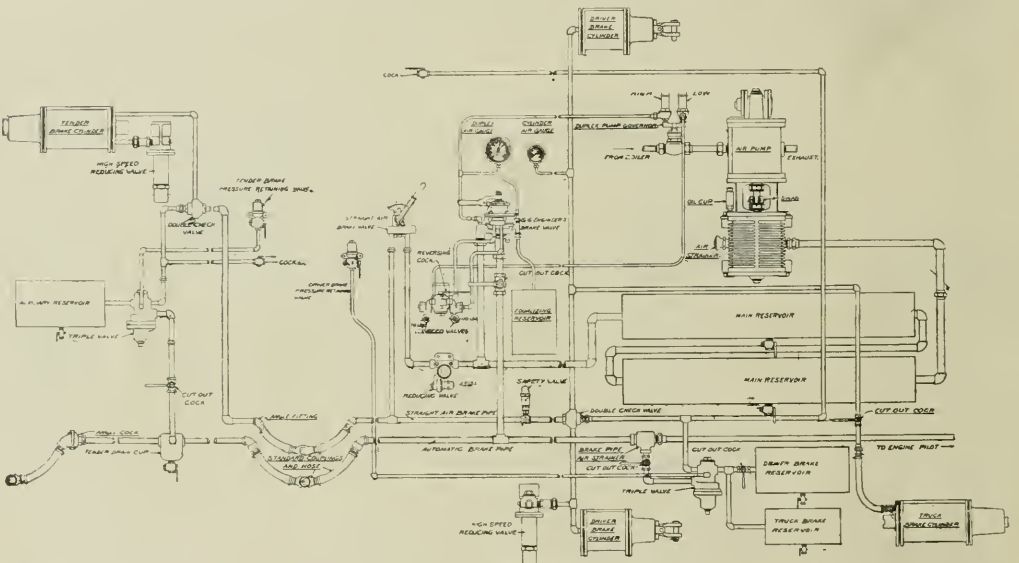


FIG. 2. DIAGRAM OF WESTINGHOUSE AIR BRAKE ARRANGEMENT, PARTS OF WHICH ARE NOW SUPERSEDED BY THE NEW "ET" EQUIPMENT.

From a glance at the picture of the parts of the old equipment displaced by the new, it will be seen that no auxiliary reservoirs and no triple valves are used with the latter. They are not necessary, since main reservoir air is used in all of the brake cylinders to apply the brakes, and the distributing valve being automatic in its action during the time the brake valve handle is on lap, keeps the brake cylinders supplied with air to the required pressure, regardless of leakage — something the triple valves do not do.

Length of piston travel or its variation makes no difference in the cylinder pressure developed, it being the same in all the brake cylinders on the engine

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Value of Reading.

Herschell, the eminent astronomer, who was accustomed to make use of accurate knowledge, said: "Were I to pray for a taste which should stand me in good stead under every variety of circumstances and be a source of happiness and cheerfulness to me during life, it would be a taste for reading." He spoke wisely, and had he lived in our day, he would have seen how easily the desire for reading can be gratified, how the garnered wisdom of the ages, as crystallized in the printed page, is within the reach of the humblest. He also would have seen the growing necessity for reading in order that one may not only keep abreast of the times, but also continue to be instructed in the new and complex changes that occur in every department of human endeavor. In no sphere of industrial activity is this more true than in that of railroading. Our correspondence school aims at keeping our readers instructed in all that is new in construction, as well as all that is put to new uses. In the lesson this month we present in brief form the details of the lubricator and its attachments with the assurance that the questions and answers will receive the careful attention of our readers.

Second Series of Questions.

61. Describe the manner in which a sight feed lubricator operates.

A.—Sight feed lubricators operate on the hydrostatic principle. In other words, the lubricator depends on the action of natural laws which have to be complied with or failure is the result. The positive steam pressure in the lubricator and pipes being the same as the back pressure, these soon neutralize each other and this leaves the positive pressure of the column of water extending through the pipe leading from the condenser, the condenser itself, and the part of the pipe above it; against this is opposed the back pressure of the column of water in the sight feed glass. As the latter is much less than the positive pressure, the drops of oil are forced through the nozzle.

62. How should a lubricator be shut off before filling?

A.—The feed valves should be shut off first, the water valves next and the steam valves last.

63. Will any bad results ensue from filling a lubricator full with cold oil?

A.—Yes; if the lubricator is not provided with an expansion chamber. Oil expands about 1/5 in volume while being heated, from about 70 degrees to the temperature of steam at 200 lbs. pressure, which is 387.7 degrees F.

64. Which is the better practice, to close the feed valves or water valve while waiting on sidings, etc.?

A.—The better practice is to close the water valve instead of the regulating valve while waiting on side tracks, or while waiting for orders. By so doing the hydrostatic pressure is cut off from the body of the lubricator and the pressure within the body of the lubricator equalizes with that in the feed glass chamber. Then, by opening the water valve, the feeds will start off at the same number of drops per minute as originally, as the adjustment has not been interfered with. It is not generally understood that if it becomes necessary to do any switching or if the engineer wishes for any reason to feed less oil, he can slow down the number of drops per minute by simply throttling the water valve. This will give the same effect as adjusting the feeds to a slower rate. The only objection that can be made to closing the water valve in these cases is that the air pump does not receive oil while the water valve is thus closed.

65. In what order should valves on lubricator be opened?

A.—First, the steam valve. Next, the water valve. Last, the regulating stems. Under no circumstances should the water valve be opened first.

66. Does the draft from the open cab window affect the working of the lubricators? Why?

A.—It did with the old tubular glass type of lubricator. The draft blowing on the lower feed arm congealed the oil to such an extent that it affected the rate of feed, but with the newer type of lubricator, such as the Detroit No. 21, the feed chamber is within the body, and the feeds are not affected by any drafts or change of temperature.

67. What else might cause irregularity of the feed?

A.—There are three conditions. First, if the engineer should throttle the steam to the lubricator down to that point where it becomes wire-drawn, the combined steam chest pressures would overcome the lubricator pressure and the feed would slow down. Second, if sediment should gather in the water passage to such an extent that the openings would be closed up, this condition would be similar to throttling the water valve and would reduce the number of drops per minute. Third, when the opening through the small choke in the automatic steam chest plug becomes enlarged beyond 7/64 in. the rate of feed will vary. The same will happen when the opening through the chokes at the lubricator end of the oil pipe in some types of lubricators becomes enlarged.

68. If the lubricator feeds faster when the throttle is closed than open, where is the trouble?

A.—In the chokes. [In this connection the reader is referred to some remarks on automatic steam chest plugs and valves which are to be found on page 12 of the pamphlet issued by the Detroit Lubricator Company, Hodges Building, Detroit, Mich., a copy of which may be obtained by direct application to the company.—Ed.]

69. If the sight feeds get stopped up, how should they be cleaned?

A.—This question has reference to lubricators while in service filled or partly filled with oil. Cleaning can be accomplished by either of the following methods: First, open the feed regulating valve wide to the feed in question and open the sight feed drain stem. The result will be a lessened pressure in the sight feed chamber which will force the clogging matter out of the cone. If the obstruction is too great to be forced out in this way try closing the other regulating feed stems and also the water valve. Open the drain valve to the body of the lubricator. This will allow the pressure to escape through the body of the lubricator and at the same time the steam pressure will blow down through the feed cone and will dislodge the obstruction and force it into the body of the lubricator.

70. How should the choke plugs be cleaned?

A.—Take them out and examine them. Run a small wire through the opening. They should be renewed when the opening is 7/64 in. in diameter.

71. Can you explain the use of the equalizing pipe?

A.—The main object of the equalizing tubes is to overcome the back pressure from the steam chest. At the same time it furnishes condensation to the sight feed chambers. By the use of these tubes the same pressure exists on top of the water in the sight feed chamber as on the surface of the water in the condenser and this steadies the feed.

72. What will be the effect if an equalizing pipe were broken off or became very loose?

Walschaert Valve Gearing.

BY JAMES KENNEDY.

The popular favor with which the Walschaert valve gearing is being received wherever it has been recently introduced, gives indication that it is in every way better suited for the heavy modern locomotive than the shifting link motion so long in use. Some difficulties are being experienced in adjusting the gearing in case of repairs and alterations on locomotives, but the movable points about the gearing are very simple, and, as a rule, require little or no readjustment. In comparison with the Stephenson shifting link there is little variation in the action of the Walschaert gearing, whereas, as is well known, the variations in the Stephenson gearing are of rapid and complex growth.

In adjusting the Walschaert valve gearing all that is necessary is to place the engine on either of the dead centers.

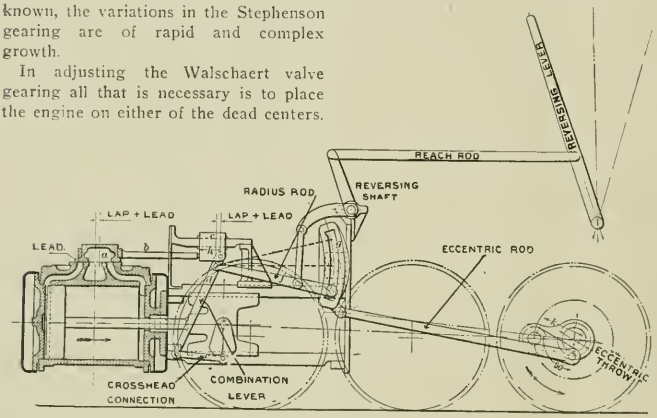


DIAGRAM OF WALSCHAERT VALVE GEAR.
(Reproduced from Baldwin Locomotive Works Catalogue.)

The length of the eccentric rod should then be so adjusted that in moving the reverse lever from end to end of the quadrant the valve should remain perfectly still in its position while the radial bar moves from end to end of the link. Any movement of the valve shows that the eccentric rod is too long or too short, as the case may be. This should be tried at both centers and if the valve remains motionless the eccentric rod is of correct length. It is well, before making any change on the length of the eccentric rod, to see that the valve rod is of the proper length. This can readily be determined by marking the position of the valve at both dead centers and lengthening or shortening the valve spindle to equalize variations, if any. The amount of lap or lead is then determined by the leverage in the combination attached to the crosshead. It will be readily seen that by reason of the eccentric crank being set at right angles to the main crank, and the eccentric rod and valve connections being of the proper length, the slide valve, unattached to the crosshead connection, would be in the center of the valve seat with, perhaps, the little divergence from the exact center arising from the slightly angular advance of the eccentric rod. Assuming that the valve is on the center the distance to be moved amounts to whatever the lap and lead may be, and the proportion of the combination lever with the short leverage connection with the crosshead can readily be determined by moving the valve to its proper position, and then with the valve spindle and radial bar coupled, the length of the lower connection from the crosshead to the combination lever can be readily determined.

It will thus be seen that if the eccentric rod and connecting rod between the crosshead and combination or lead lever, as it might very properly be called, were made adjustable by substantial turnbuckles or other movable devices, the setting of the Walschaert valve gearing and the retaining of the valve in its proper position would be a matter of easy accomplishment. It may be noted that a locomotive constructed at Taunton, Mass., and put in operation at the Centennial Exhibition, at Philadelphia, in 1876, with the Walschaert gearing moving the valves, was fitted with an adjustable turnbuckle on the eccentric rod, while another locomotive, exhibited at St. Louis in 1903, has a similar device attached to the crosshead connection with the lead lever. It is to be hoped that our locomotive builders will soon produce a locomotive with eccentric rods, valve rods and crosshead connections all readily and securely adjustable in length.

Meanwhile, in adjusting the eccentric rod, it has become a practice in some shops to change the position of the bearings suspending the link as in the case of the eccentric rod being too short a liner between the guide yoke, or bearing brace, and link bearing casting will shorten the distance between the oscillating link and eccentric crank. This has the pernicious effect of disturbing the true location of the radial bar and paves the way for organic defects, and should be avoided for no other reason than that liners never fail to lend themselves to the loosening of substantial bearings, and one of the best features of the Walschaert valve gearing is the opportunity that the parts afford for broad and massive bearings which, with the lessened number of working joints, give the apparatus a most desirable degree of stability.

It may be added that in adjusting valves with the Walschaert gearing while it is well to begin the operation with the reverse lever at the extreme ends of the quadrant, it is necessary to finish the work by careful experiment at the point of cut-off at which the locomotive will be expected to do the greater part of its work, for, while the lead remains a fixed quantity, it will be found that there are causes which produce variations in all mechanical movements that have their origin in circles or parts of circles and are transferred to a reciprocating or lineal motion.

One of the most conspicuous defects noted in the operating of locomotives during the past winter has been steam blowing through stuffing box glands. Besides being a serious waste of steam the leakage is a dreaded source of danger since it frequently obscures the track ahead and the signals that conserve safety. We have heard frequent discussions of the matter by the men responsible and they have generally devoted their attention to methods of improving the packing. The fact is that the packing is nearly as perfect as it can be made and that the real defect is in the rods that work in the packing. They are roughly turned and tear the packing from the start. Railroad men ought to devote more attention to the finish of the rods. Those who use piston and valve stem rods finished on machines made by the Norton Grinding Co., of Worcester, Mass., will find very little annoyance from leaky packing.

Greene, Tweed & Co., the makers of Palmetto air pump packing, have moved into more commodious quarters on account of the steady increase of business. Their new location is at 109 Duane street, New York City.

The Chinese have many moral maxims that might be profitably introduced into American practice, if that would influence public morals. Yong Chen, one of their philosophers, when asked to take a bribe, and assured that no one would know it, answered: "How so? Heaven would know, Earth would know, you would know, and I should know."